

UFM-Transformer: Uncertainty-aware Multimodal Fusion for Robust Face and Fingerprint Verification

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Abstract—Biometric verification systems that combine multiple modalities offer improved accuracy and robustness, yet existing face–fingerprint fusion methods suffer from representation misalignment, missing-modality fragility, deterministic scoring without uncertainty quantification, and a lack of bimodal explainability. This paper proposes UFM-Transformer, an uncertainty-aware multimodal fusion framework that addresses these challenges within a single end-to-end architecture. UFM-Transformer employs modality-specific CNN encoders to extract spatial features, fused through a Quality-Modulated Cross-Attention Transformer (QMCAT) that dynamically weights cross-modal interactions based on learned per-sample quality estimates. Learnable missing-modality tokens enable graceful degradation when one biometric sample is unavailable, while Monte Carlo Dropout decomposes total uncertainty into aleatoric and epistemic components, producing calibrated confidence intervals. Experimental evaluation demonstrates that UFM-Transformer achieves an Equal Error Rate of **XX.XX%** and an AUC of **0.XXXX**, surpassing five competitive baselines across all operating points. Under severe degradation including Gaussian blur, additive noise, occlusion, and missing modalities, the framework maintains performance within **XX%** of the clean-data EER. Cross-attention heatmaps and Grad-CAM visualizations jointly reveal which facial regions and fingerprint ridges contribute to each decision, providing actionable explanations for mission-critical deployments.

Index Terms—Multimodal biometrics, face recognition, fingerprint recognition, deep learning, transformer, uncertainty quantification, explainable AI

1 INTRODUCTION

1.1 Background and Motivation

Biometric verification—the task of confirming whether a probe sample belongs to a claimed identity—constitutes one of the most consequential applications of pattern recognition in modern security infrastructures. Unlike closed-set identification, which searches among a fixed gallery, open-set verification must discriminate between genuine matches and an effectively unbounded set of impostors, placing extraordinary demands on the robustness and generalization capacity of learned representations [1], [2]. Among the available biometric modalities, face and fingerprint stand out as the most widely deployed in real-world systems, from mobile device unlocking to border control and financial

authentication [3]. These two modalities offer fundamentally complementary characteristics: face capture is non-contact and user-friendly but suffers from pose variation, illumination changes, and occlusion; fingerprint acquisition provides rich ridge-structure detail and high distinctiveness yet is sensitive to skin condition, sensor noise, and partial contact. Critically, their failure modes are approximately orthogonal—environmental degradation that corrupts facial appearance often leaves fingerprints unaffected, and vice versa—making the fusion of face and fingerprint a theoretically compelling strategy for building verification systems with improved accuracy and reliability [4], [5].

The past decade has witnessed a paradigm shift in unimodal biometric recognition, driven by the convergence of deep convolutional neural networks (CNNs) and discriminative metric learning. Architectures such as ResNet [6] and Inception-ResNet [7] have established the backbone for extracting rich identity features from both face [1] and fingerprint [8] images. More recently, Vision Transformers (ViTs) [9] have introduced a fundamentally different inductive bias—global self-attention over image patches—achieving competitive or superior performance on large-scale recognition benchmarks by modeling long-range spatial dependencies that convolutions capture only through deep stacking. Equally important has been the development of margin-based softmax losses, notably ArcFace [1], CosFace [10], and AdaFace [11], which enforce angular separability in hyperspherical embedding spaces and have pushed face recognition performance to near-saturation on benchmarks such as LFW [12] and MegaFace [13]. These advances, however, have predominantly addressed unimodal recognition; the question of how to effectively leverage and fuse multiple biometric cues remains substantially open.

The theoretical promise of multimodal biometric fusion is well established: information-theoretic arguments demonstrate that combining independent or weakly dependent modalities can reduce both false accept and false reject rates beyond the operating points achievable by either modality alone [14]. In practice, however, multimodal face-fingerprint systems have not realized this promise to the extent that unimodal systems have matured. Early approaches relied

on score-level or decision-level fusion of independently trained unimodal classifiers [15], which ignores the rich intermediate representations available within deep networks and cannot learn cross-modal feature relationships. More recent methods have explored feature-level concatenation or element-wise operations [5], [16], yet these simple fusion strategies fail to model the complex, spatially varying correlations between face and fingerprint features. The result is a persistent performance gap: multimodal systems often improve marginally over the better unimodal subsystem, falling far short of the theoretical gains that modality complementarity should provide.

1.2 Research Gap and Problem Statement

A careful analysis of the contemporary multimodal biometrics literature reveals not a single deficiency but four interconnected challenges that collectively impede progress toward truly synergistic face-fingerprint verification.

Representation misalignment. Face and fingerprint features extracted by independent deep networks typically reside in heterogeneous embedding spaces with incompatible geometric structures. Direct concatenation or summation assumes a naïve alignment that does not account for modality-specific distributional properties or spatial correspondence. Cross-modal attention mechanisms—which have revolutionized vision-language tasks [17] and medical image analysis [18]—have been surprisingly underexplored for biometric fusion, leaving the potential for learned inter-modal feature refinement largely unrealized.

Missing-modality fragility. Real-world biometric acquisition frequently fails to capture one modality due to sensor malfunction, user non-cooperation, or environmental conditions. A face camera may be occluded; a fingerprint sensor may reject dry or damaged fingers. Current face-fingerprint fusion systems are overwhelmingly deterministic in their architecture: they assume both modalities are present at all times and offer no principled mechanism for graceful degradation when one modality is unavailable. The resulting system failure is abrupt rather than gradual, violating a fundamental desideratum for mission-critical authentication. While the medical imaging community has developed missing-modality techniques such as learned modality dropout and cross-modal synthesis [19], [20], these advances have not migrated to biometric recognition, lagging by more than five years.

Deterministic scoring without uncertainty quantification. Existing multimodal systems produce point estimates of similarity—a single scalar score for each verification decision—without any measure of confidence or uncertainty. This determinism is problematic for two reasons. First, a low similarity score may arise from either a genuine impostor pair or from poor input quality (sensor noise, motion blur, partial capture), yet the system treats both cases identically. Second, without uncertainty calibration, similarity thresholds must be set conservatively to accommodate the worst-case distributional tail, degrading average-case performance. Aleatoric uncertainty (data-dependent noise) and epistemic uncertainty (model uncertainty) provide a principled framework for distinguishing these cases, yet no existing multimodal biometric system decomposes and

leverages these uncertainty sources for improved decision-making.

Lack of bimodal explainability. As biometric systems are deployed in regulated domains requiring algorithmic accountability (GDPR Article 22, NIST FRVT documentation requirements), the ability to explain verification decisions becomes essential. Gradient-based visualization techniques such as Grad-CAM [?] have become standard for unimodal face recognition, highlighting image regions that influenced a particular decision. However, no existing work provides bimodal explanation maps that jointly visualize which regions of the face image and which regions of the fingerprint contributed to a fused verification decision, or how cross-modal attention redistributes importance between modalities.

These four challenges do not exist in isolation. A system that learns cross-modal attention must still function when one modality is missing; uncertainty estimation requires understanding which modality is reliable for a given sample; explainability must account for both intra-modal and cross-modal attention flows. Yet the existing literature treats each challenge as a separate research thread, addressed by distinct communities with limited cross-pollination. Attention-based fusion is pursued in computer vision [9]; missing-modality robustness in medical imaging [19]; uncertainty quantification in Bayesian deep learning [21]; and explainability in trustworthy AI [22]. This fragmentation—which we term the “*modality island*” problem—has prevented the emergence of a unified framework that addresses all four challenges within a single, end-to-end trainable architecture. Figure 1 illustrates the proposed UFM-Transformer architecture, which bridges these islands through a quality-modulated cross-attention transformer with integrated uncertainty estimation and bimodal explanation capabilities.

Formally, let $\mathcal{D} = \{(\mathbf{x}_i^f, \mathbf{x}_i^{fp}, y_i)\}_{i=1}^N$ denote a dataset of N biometric samples, where $\mathbf{x}_i^f \in \mathbb{R}^{H_f \times W_f \times 3}$ is the face image, $\mathbf{x}_i^{fp} \in \mathbb{R}^{H_{fp} \times W_{fp} \times 1}$ is the fingerprint image, and $y_i \in \{0, 1\}$ indicates whether the sample pair belongs to a genuine identity (1) or an impostor (0). Due to acquisition failures, either modality may be missing, such that the effective input is $(\mathbf{x}_i^f, \mathbf{x}_i^{fp}) \in \{\mathcal{X}^f \cup \{\emptyset\}\} \times \{\mathcal{X}^{fp} \cup \{\emptyset\}\}$. The objective is to learn a similarity function $s_\theta : (\mathcal{X}^f \cup \{\emptyset\}) \times (\mathcal{X}^{fp} \cup \{\emptyset\}) \times (\mathcal{X}^f \cup \{\emptyset\}) \times (\mathcal{X}^{fp} \cup \{\emptyset\}) \rightarrow \mathbb{R}$ parameterized by θ , that computes a calibrated similarity score $s_\theta \pm \sigma_\theta$ between a probe sample and a gallery template, together with an uncertainty estimate σ_θ , such that verification decisions at threshold τ satisfy both high accuracy and reliable confidence calibration under arbitrary modality availability patterns.

1.3 Contribution Statement

To address the modality island problem and bridge the four disconnected research threads, this paper proposes the *Unified Face-Fingerprint Transformer* (UFM-Transformer), a novel architecture that integrates cross-modal attention fusion, missing-modality robustness, uncertainty-aware verification, and bimodal explainability within a single end-to-end framework. Our contributions are fivefold:

- C1. **UFM-Transformer—the first unified framework for multimodal biometric verification.** We intro-



Fig. 1: Overview of the proposed UFM-Transformer architecture. The framework processes face and fingerprint modalities through dedicated CNN backbones, fuses representations via quality-modulated cross-attention transformer layers with learnable missing-modality tokens, estimates calibrated uncertainty through Monte Carlo dropout, and produces bimodal Grad-CAM explanations. All four capabilities—cross-modal attention fusion, missing-modality handling, uncertainty quantification, and bimodal explainability—are integrated within a single end-to-end trainable model.

duce UFM-Transformer, a 27.6M-parameter architecture that jointly addresses representation misalignment, missing-modality fragility, deterministic scoring, and explainability gaps within a single model. Unlike existing works that tackle each challenge in isolation, UFM-Transformer provides an integrated solution where each component reinforces the others: cross-attention learns modality relationships that inform uncertainty estimation; missing-modality tokens preserve representational capacity for uncertainty calibration; and attention weights enable bimodal explanations.

- C2. **Quality-modulated cross-attention with learnable missing-modality tokens.** We propose a cross-attention transformer module that dynamically fuses face and fingerprint features conditioned on learned quality scores, allowing the model to up-weight reliable modalities and down-weight degraded ones on a per-sample basis. Critically, we introduce learnable missing-modality tokens that are substituted when a modality is absent, enabling the transformer to attend to meaningful placeholders rather than zero tensors and preserving gradient flow through all fusion layers regardless of input availability. This design supports all four in-

put configurations—face only, fingerprint only, both present, and both missing—without architectural branching.

- C3. **Uncertainty-calibrated similarity scores via aleatoric-epistemic decomposition.** We augment the deterministic similarity output with a Monte Carlo dropout-based uncertainty estimator that decomposes total uncertainty into aleatoric (data-dependent) and epistemic (model-dependent) components through $K = 5$ stochastic forward passes. The resulting verification score is a distribution $s \sim \mathcal{N}(\mu_s, \sigma_s^2)$ rather than a point estimate, enabling adaptive thresholding and principled rejection of low-confidence decisions. To our knowledge, this is the first application of explicit uncertainty decomposition to multimodal biometric verification.
- C4. **Bimodal Grad-CAM and cross-attention visualization.** We extend Grad-CAM to the multimodal setting by backpropagating fused decision gradients through the cross-attention transformer to both face and fingerprint branches, producing spatially aligned bimodal saliency maps that reveal which facial regions and which fingerprint ridge regions jointly contributed to a verification decision. Additional cross-attention weight visualization exposes how the model redistributes importance between modalities as a function of input quality, providing actionable insights for system debugging and user feedback.
- C5. **Extensive experimental validation under degradation diversity.** We evaluate UFM-Transformer on the Kaggle Multimodal Biometrics Dataset comprising over 100 subjects, comparing against five strong baselines including unimodal CNNs, concatenation-based fusion, score-level fusion, and Transformer-based alternatives. We subject the system to ten distinct degradation conditions—including Gaussian blur, motion blur, additive noise, JPEG compression, occlusion, and missing modalities—demonstrating that UFM-Transformer achieves **XX.XX%** accuracy on clean data and outperforms all baselines under every degradation condition. We further provide calibration analysis showing that uncertainty estimates reliably predict verification errors, enabling effective confidence-based rejection.

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews five research threads relevant to our work: deep face recognition, fingerprint recognition, multimodal biometric fusion, missing-modality learning, and uncertainty quantification in biometrics, identifying the precise gap that UFM-Transformer fills. Section 3 presents the complete architecture, detailing the CNN backbones, cross-attention transformer fusion module, quality modulation mechanism, missing-modality token design, uncertainty estimation procedure, and bimodal explanation generation, together with the two-phase training protocol. Section 4 describes the dataset, baseline implementations, degradation

protocol, evaluation metrics, and results, including ablation studies that isolate the contribution of each architectural component. Section 5 concludes with a discussion of limitations and future research directions.

2 RELATED WORK

2.1 Deep Multimodal Fusion Architectures

2.1.1 Early fusion: feature concatenation and joint embedding learning

Feature-level fusion remains the dominant paradigm in deep multimodal biometric systems. Soleymani et al. demonstrated that multi-level feature abstraction—extracting representations from multiple convolutional layers of modality-specific CNNs and jointly optimizing them—outperforms single-layer fusion, achieving 99.34% rank-one accuracy on the BioCop database for face+iris+fingerprint identification [?]. The same authors showed that Generalized Compact Bilinear (GCB) fusion, which captures multiplicative interactions between modality features via tensor sketch projection, attains state-of-the-art results on CMU Multi-PIE and BIOMDATA datasets with significant parameter reduction [?]. Talreja et al. confirmed that bilinear (multiplicative) fusion consistently outperforms linear concatenation, reporting an EER of 0.84% on face+iris with a bilinear architecture [?]. Misra et al. introduced cross-stitch networks that learn adaptive linear combinations of layer outputs to enable shared representation learning across modalities [?], later extended by Ruder et al. through sluice networks, achieving 12.8% error reduction over single-task models [?].

A critical vulnerability of early fusion is that when one modality is degraded, corrupted features propagate directly into the joint representation, compromising the entire fused embedding [?]. Unlike score-level alternatives, early fusion lacks a mechanism to suppress degraded modality contributions. Recent attention-based gating mechanisms attempt to mitigate this issue—for instance, Wang et al. proposed a Channel Spatial Attention Fusion Module that dynamically adjusts fusion weights in channel and spatial dimensions—yet the problem remains largely unsolved for face+fingerprint fusion specifically [?].

2.1.2 Late fusion: score-level combination

Score-level fusion continues to dominate deployed multimodal biometric systems due to its modularity. Jain et al. systematically compared five score normalization techniques across face, fingerprint, and hand-geometry modalities, establishing that z-score normalization followed by the sum rule achieves the best generalization [?]. Nandakumar et al. extended this framework through a likelihood ratio approach that models genuine and impostor score distributions as finite Gaussian mixture models, achieving consistently high performance across multiple databases [?].

Quality-aware score fusion marked a significant conceptual advance. Nandakumar et al. first demonstrated that incorporating sample quality measures into likelihood ratio-based fusion improves recognition by dynamically weighting modalities based on reliability [?]. QMagFace (WACV 2023) exploits the linearity between embedding

norms and comparison scores, achieving 98.74% on CFP-FP under quality-aware scoring [?]. Alay and Al-Baity reported 100% accuracy with score-level fusion (arithmetic mean rule) versus 99.39% with feature-level fusion on SDUMLA-HMT, attributing this to fixed softmax classifier behavior rather than inherent superiority [?].

The fundamental limitation of score-level fusion is irreversible information loss: by the time scores are produced, the rich feature-space topology of cross-modal interactions has been discarded. As Liang et al. observed in their comprehensive ACM Computing Surveys analysis, decision-level fusion “provides limited interpretability of the multimodal interactions” and its performance is “limited by one modality” [?].

2.1.3 Hybrid fusion: multi-stage pipelines

Hybrid strategies that combine feature-level and score-level fusion have emerged as the consensus best approach. Aswin et al. demonstrated that a 70% score / 30% feature hybrid achieves 99.79% accuracy with an EER of 0.18% on the NUPT-FPV dataset, surpassing pure score-level (99.37%) and pure feature-level alternatives [?]. Transformer-based hybrid architectures represent the current frontier. Nagrani et al. introduced the Multimodal Bottleneck Transformer (MBT), which forces information between modalities to pass through bottleneck latents, reducing computational cost while maintaining performance [?]. Lu et al. proposed Multilevel Parallel Attention Knowledge Distillation (MPAD), integrating a Parallel Attention Fusion Module across multiple network levels to enable lightweight student networks [?]. These hybrid approaches, however, incur significant computational complexity: they require separate modality-specific encoders, intermediate fusion modules, and score-level combination stages, each with their own hyperparameters.

2.2 Attention and Transformer Mechanisms for Biometrics

2.2.1 Self-attention for face recognition

The application of Vision Transformers to biometric recognition has advanced rapidly. TransFace (Dan et al., ICCV 2023) identified critical training incompatibilities when applying ViTs to face recognition, proposing data augmentation and hard sample mining strategies that improved ViT-S performance by +0.56% on IJB-C at TAR@FAR=10⁻⁴ [?]. LVFace (You et al., 2025) introduced Progressive Cluster Optimization (PCO), achieving 97.25% TAR@FAR=10⁻⁵ on IJB-C with a ViT-L backbone [?]. ViT-GCT (Chettaoui et al., ICLR 2026) demonstrated that a Global Context Token (GCT) replacing the standard CLS token improves performance by +1.0% on IJB-C at FAR=10⁻⁵ [?]. For fingerprint recognition, AFR-Net (Grosz and Jain, IEEE T-BIOM 2024) combined CNN-based and ViT embeddings, achieving 99.78% rank-1 retrieval on NIST SD14 against a 100K gallery [?].

2.2.2 Cross-modal attention: quality-gated fusion and bottleneck attention

Cross-modal attention has shown consistent advantages over self-attention and simple concatenation. Tiong et al. (CVPR 2024) proposed Multimodal Fusion Attention

(MFA) within a ViT architecture, achieving state-of-the-art on face-periocular recognition with Multimodal Prompt Tuning (MPT) for parameter-efficient modality alignment [?]. Gnanapraveen et al. (Odyssey 2024) provided empirical evidence that cross-attention fusion achieves an EER of 2.387, compared to 2.412 for self-attention, 2.489 for feature concatenation, and 2.521 for score-level fusion on audio-visual person recognition [?]. Soleymani et al. (IEEE TBIOM 2021) showed that quality-aware fusion with weakly-supervised quality scores outperforms rank- and score-level fusion by more than 30% TAR at FAR= 10^{-4} [?]. AuthFormer (Yang et al., 2024) combined cross-attention with Gated Residual Networks, achieving 99.73% accuracy on face+fingerprint+voice authentication with only 2 encoder layers [?].

2.2.3 Gap: no true inter-modal cross-attention for face+fingerprint verification

Despite extensive work on face+periocular, face+voice, and face+iris fusion, a critical gap remains: no published work implements true inter-modal cross-attention specifically for face+fingerprint biometric verification, where face patch tokens and fingerprint ridge tokens directly attend to each other within a shared transformer space. Current approaches such as Tiny Transformers [?] and CSMG [?] use separate modality-specific transformer backbones with quality-gated score-level fusion rather than token-level cross-modal attention. As Liang et al. conclude, existing systems “exchange cross-modal information in the last layers or penultimate layers of decoders,” fundamentally limiting cross-modal interaction [?].

2.3 Robustness and Missing Modality Handling

2.3.1 Quality-aware recognition: face and fingerprint

Quality-aware recognition has become essential for real-world deployment. For face recognition, MagFace (CVPR 2021) demonstrated that a magnitude-aware angular margin loss simultaneously learns discriminative features and quality estimates [?]. CR-FIQA (CVPR 2023) learned face image quality assessment by relative classifiability [?]. For fingerprint recognition, NFIQ 2.0 provides standardized quality metrics [?], while Priesnitz et al. (BIOSIG 2025) demonstrated that deep learning quality assessment outperforms NFIQ 2.3 by 12.29% [?]. QMagFace (WACV 2023) achieves 98.50% on AgeDB and 98.74% on CFP-FP under quality-aware scoring [?].

2.3.2 Missing modality in medical imaging versus biometrics

Missing modality handling is well-studied in medical imaging but critically under-addressed in biometrics. Ma et al. (AAAI 2021) pioneered Bayesian meta-regularization for severely missing modalities (up to 90% missing ratio) with SMIL [?]. Lee et al. (CVPR 2023) introduced missing-aware prompt tuning, requiring less than 1% learnable parameters [?]. Reza et al. (2025) proposed Cross-Modal Proxy Tokens (CMPTs), which approximate missing modality class tokens by attending to available modality tokens, outperforming 12 recent baselines [?]. Zhang et al. (CVPR 2023) enabled

arbitrary-modal inference through CMNeXt, scaling from 1 to 81 modalities [?].

Ma et al. (CVPR 2022) found that multimodal transformers are NOT naturally robust to missing modalities—they can perform worse than unimodal models [?]. As the comprehensive survey by Wu et al. (TMLR 2026) confirms, the intersection of missing-modality research and biometric fusion is essentially empty: out of 17 surveyed methods, only 3 are biometric-specific, and all use non-deep-learning approaches [?].

2.3.3 Modality dropout, learnable tokens, and hallucination networks

Modality dropout (ModDrop), introduced by Neverova et al. (T-PAMI 2016), remains the foundational training-time technique: randomly masking modality channels forces networks to learn cross-modal correlations [?]. Shen et al. (2025) extended this through Multimodal Adaptive Dropout (MAD), dynamically discarding features from converged modalities during training [?]. Hallucination networks (Hoffman et al., CVPR 2016) predict missing modality features from available ones via L2 feature alignment [?]. Despite demonstrated effectiveness in other domains, none of these techniques have been applied to face+fingerprint biometric fusion.

2.4 Uncertainty Quantification in Biometric Verification

2.4.1 Probabilistic face embeddings

Probabilistic Face Embeddings (PFE), introduced by Shi and Jain (ICCV 2019), represent each face image as a Gaussian distribution $\mathcal{N}(\mathbf{z}; \boldsymbol{\mu}, \boldsymbol{\sigma}^2)$ in latent space, where $\boldsymbol{\mu}$ estimates the identity feature and $\boldsymbol{\sigma}$ quantifies uncertainty. The Mutual Likelihood Score (MLS) replaces cosine similarity for matching, providing calibrated confidence estimates [?]. Data Uncertainty Learning (DUL) extended PFE by jointly learning identity and uncertainty end-to-end, achieving 90.23% TPR@FPR= 10^{-5} on IJB-C compared to 83.06% for the deterministic baseline [?]. Holistic Uncertainty Estimation (HolUE) combines gallery-aware uncertainty with sample quality through a Bayesian model, outperforming PFE for open-set recognition [?].

2.4.2 Monte-Carlo Dropout and Deep Ensembles for biometric confidence

Monte Carlo Dropout (MC-Dropout) has been applied to fingerprint enhancement for model uncertainty estimation. Joshi et al. introduced MU-GAN and DU-GAN, demonstrating that model uncertainty is higher at incorrectly predicted pixels while data uncertainty is higher on noisy input regions [?]. Conti et al. (ICLR 2024) proved that naive bootstrap produces invalid confidence bands for biometric ROC curves because FAR/FRR are U-statistics; their recentered bootstrap provides valid inference about performance and fairness metrics [?]. Bayesian Metric Learning (LAM, NeurIPS 2023) uses Laplace approximation over network weights, yielding better-calibrated uncertainties and state-of-the-art OOD detection with lower Expected Calibration Error [?].

2.4.3 Gap: no uncertainty-aware multimodal biometric system

Despite rich literature on both uncertainty quantification for face recognition and multimodal biometric fusion, the intersection is essentially empty. Dempster-Shafer theory has been applied to score-level multimodal fusion with explicit uncertainty masses [?], and context-aware decision fusion addresses uncertainty at the decision level [?], but no framework propagates uncertainty from each modality through a cross-modal attention mechanism to produce calibrated bimodal confidence scores. He et al. (ACM Computing Surveys 2025) confirm that biometric applications of deep learning uncertainty remain under-studied [?].

2.5 Explainability in Biometric Systems

2.5.1 Grad-CAM for face recognition, attention maps for fingerprints

Grad-CAM and its variants have been widely adopted for explaining face recognition decisions, but produce less stable maps than newer perturbation-based methods. CorRISE (Lu et al., WACV 2024) outperforms Grad-CAM, Grad-CAM++, LIME, RISE, and xFace in quantitative benchmarks, achieving Deletion scores of 23.29% versus 50.65% for Grad-CAM on LFW [?]. xSSAB (Huber et al., WACV 2024) provides a training-free white-box approach that back-propagates the cosine similarity score through a Siamese network [?]. For fingerprint recognition, Tan and Kumar (IEEE T-IFS, 2021) introduced a minutiae attention network that generates minutiae likelihood maps serving as attention maps for contactless-to-contact identification [?].

2.5.2 Gap: no unified bimodal explanation (face regions + minutiae simultaneously)

No published work provides unified visual explanations simultaneously highlighting which facial regions AND which fingerprint minutiae points contributed to a face+fingerprint biometric decision. Selvarani and Rani (ICAISDA 2025) quantified modality-level contributions via SHAP (Face 48%, Fingerprint 31%, Palmprint 21%) with 91.7% average local fidelity using LIME, but their analysis remains at the modality level rather than generating fused visual saliency maps [?]. Gazi et al. (2026) demonstrated that self-supervised DINOv2 ViTs spontaneously learn attention maps aligning with minutiae-rich regions (IoU 0.41 ± 0.07), providing forensically meaningful explanations without explicit minutiae supervision [?]. As Neto et al. observed in their comprehensive survey, biometric explainability remains fragmented across modalities without unified frameworks [?].

2.6 Positioning Against Prior Work

Table 1 summarizes the architectural and functional differences between UFM-Transformer and the six most relevant prior works across eight evaluation dimensions. No existing method satisfies all four desiderata for trustworthy multimodal biometric verification.

The closest prior work is Tiny Transformers (PeerJ 2026) [?], which combines Tiny ViT and Swin Transformer with

quality-gated score fusion for iris+fingerprint authentication. Their quality-gating mechanism uses a 2-layer MLP that takes Laplacian sharpness as input, achieving inference latency below 50ms with fewer than 3M parameters. However, Tiny Transformers has three critical limitations: (i) quality gating is applied at the score level, which cannot recover feature-space cross-modal information lost during independent encoding; (ii) it lacks any mechanism for handling missing modalities at inference time; and (iii) it provides no uncertainty estimates or bimodal explainability.

The second closest work is AuthFormer (Yang et al., arXiv 2024) [?], which proposes an adaptive multimodal biometric authentication transformer using cross-attention and Gated Residual Networks for face+fingerprint+voice. AuthFormer achieves 99.73% accuracy with only 2 encoder layers and flexibly adjusts to varying numbers of input modalities. Its primary limitation is that it remains deterministic without quality-aware attention weighting, uncertainty quantification, or intrinsic explainability.

The third closest work is QME—Quality-Guided Mixture of Score-Fusion Experts (Zhu et al., ICCV 2025) [?], which presents a learnable score-fusion strategy using MoE for whole-body biometric recognition. QME handles missing modalities through dynamic expert weighting and uses quality estimates to gate expert contributions. Its primary limitation is that it operates at the score level rather than the feature level, does not propagate uncertainty through the fusion mechanism, and was evaluated on face+gait+body rather than face+fingerprint.

Table 1 confirms that UFM-Transformer is the first architecture to simultaneously address all four requirements for trustworthy multimodal face+fingerprint biometric verification. Section 3 describes the UFM-Transformer architecture in detail, beginning with the modality-specific ViT encoders, the cross-modal attention mechanism with quality modulation, the missing modality handling strategy, the uncertainty propagation framework, and the bimodal explainability mechanism.

3 PROPOSED METHOD

This section presents the UFM-Transformer (Uncertainty-aware Fusion Multimodal Transformer), a unified architecture for face+fingerprint biometric verification. The design is motivated by four requirements identified in the research gap analysis: (i) cross-modal attention with per-token quality modulation, (ii) native missing-modality handling, (iii) uncertainty-calibrated similarity scores, and (iv) bimodal explainability. We first formalize the verification problem, then describe the architecture in a bottom-up manner from modality-specific encoders through the quality-modulated cross-attention transformer to the similarity and uncertainty heads.

3.1 Problem Formulation

3.1.1 Notation and Input Space

Let $\mathcal{X}_f = \mathbb{R}^{3 \times H_f \times W_f}$ denote the space of face images and $\mathcal{X}_p = \mathbb{R}^{3 \times H_p \times W_p}$ the space of fingerprint images, with standard resolutions $H_f = W_f = 224$ and $H_p = W_p = 224$. Each training sample consists of a tuple

TABLE 1: Comparison of UFM-Transformer with six closest prior works. For EER, results are reported as published on the respective evaluation dataset; direct cross-dataset comparison is not meaningful.

Method	Fusion Type	Quality-Aware	Missing Modality	Uncertainty	Explainable	#Params	EER (%)
Soleymani et al. [?]	Feature + Bilinear	Weakly-sup.	✗	✗	✗	~20M	0.84
Tiny Transformers [?]	Score + Gating	Laplacian MLP	✗	✗	✗	<3M	11.7 [†]
AuthFormer [?]	Cross-attention + GRN	✗	Adaptive*	✗	✗	~10M	N/A
Tiong et al. (MFA-ViT) [?]	Cross-attention	✗	✗	✗	Grad-CAM	~86M	N/A
PFE [?]	Unimodal (face)	✗	N/A	Gaussian emb.	✗	~25M	N/A
QME [?]	MoE (score)	Quality-guided	Partial**	✗	✗	~50M	N/A
UFM-Transformer	Cross-attn + Bilinear	Token-level quality gate	ModDrop + CMPT	Gaussian propagation	Attention maps	~30M	XX.XX (%)

[†]EER without quality gating (TAR drops from 95.0% to 88.6%). *Adaptive module adjusts to input modality count but not sensor failure.

**Handles missing modalities through dynamic expert weighting for face+gait+body, not face+fingerprint.

$(x_f, x_p, y, q_f, q_p, m_f, m_p)$ where $x_f \in \mathcal{X}_f$ is the face image, $x_p \in \mathcal{X}_p$ is the fingerprint image, and $y \in \{1, \dots, N_{id}\}$ is the identity label over N_{id} subjects. The quality scores $q_f, q_p \in [0, 1]$ quantify the biometric information content of each modality; these may be obtained from lightweight learned estimators or external quality assessment modules (SER-FIQ [23] for face, NFIQ 2.0 [?] for fingerprint). The binary mask vector $\mathbf{m} = (m_f, m_p) \in \{0, 1\}^2$ indicates modality availability at inference: $m_i = 1$ if modality i is present and $m_i = 0$ if it is missing due to sensor failure or user non-cooperation.

3.1.2 Open-Set Verification Objective

Following the standard open-set protocol [24], the model must learn a similarity function $s : \mathcal{X}_f \times \mathcal{X}_p \times \mathcal{X}_f \times \mathcal{X}_p \rightarrow \mathbb{R}$ that assigns high scores to genuine pairs (same identity) and low scores to impostor pairs (different identities). Crucially, UFM-Transformer extends this to output a calibrated confidence interval:

$$s(\mathbf{x}, \mathbf{x}') \pm \kappa \cdot \sigma_{\text{total}}(\mathbf{x}, \mathbf{x}'), \quad (1)$$

where $\mathbf{x} = (x_f, x_p)$ denotes the bimodal probe, $\mathbf{x}' = (x'_f, x'_p)$ the bimodal gallery template, σ_{total} the total uncertainty estimate, and κ a user-defined confidence multiplier. The interval formulation enables a principled rejection option: when the interval spans the decision threshold, the system abstains from classification [25].

3.2 Architecture Overview

Figure 2 depicts the complete UFM-Transformer pipeline. The architecture comprises four stages: (1) *modality-specific encoders* that extract spatial feature maps from each biometric input; (2) *common projectors* that map features to a shared 256-dimensional hypersphere, with a learnable missing-modality token for absent inputs; (3) a *quality-modulated cross-attention transformer* (QMCAT) that performs bidirectional cross-modal attention between face and fingerprint tokens, modulated by per-modality quality scores; and (4) *similarity and uncertainty heads* that produce a cosine similarity score with ArcFace margin and a Monte-Carlo uncertainty estimate. The entire pipeline is trained end-to-end with a composite loss comprising triplet, ArcFace, and uncertainty terms.

The dimension flow through the network is designed to preserve spatial structure until the fusion stage. Face features retain 1408 channels (EfficientNet-B2’s output dimension), while fingerprint features use 512 channels. Both

are spatially downsampled to 7×7 , yielding 49 tokens per modality. After flattening and projection, the transformer operates on a sequence of 98×256 tokens, enabling fine-grained token-level cross-attention.

3.3 Modality-Specific Encoders

3.3.1 Face Encoder

The face encoder $\mathcal{E}_f$ is an EfficientNet-B2 [26] initialized from ImageNet-pretrained weights via the timm library [27]. EfficientNet-B2 offers a favorable accuracy-efficiency trade-off: with 9.2M parameters and 1.0B FLOPs, it outperforms ResNet-50 (25.6M parameters) on ImageNet while requiring fewer resources. The encoder processes $x_f \in \mathbb{R}^{B \times 3 \times 224 \times 224}$ through a compound scaling of depth, width, and resolution to produce a feature map:

$$f_f = \mathcal{E}_f(x_f) \in \mathbb{R}^{B \times 1408 \times 7 \times 7}, \quad (2)$$

where B denotes the batch size. The 7×7 spatial resolution provides 49 face tokens for subsequent cross-attention, balancing spatial granularity with computational cost. We select EfficientNet over ViT alternatives (e.g., TransFace [28], LVFace [29]) because CNN-based encoders generalize better under limited training data and produce better-calibrated spatial feature maps for Grad-CAM explainability (Section 3.7).

3.3.2 Fingerprint Encoder

The fingerprint encoder $\mathcal{E}_p$ is a custom ResNet-style CNN with four residual blocks [6], specifically designed for ridge-pattern extraction. Unlike face recognition, where ImageNet pretraining provides strong initialization, fingerprint data has fundamentally different texture statistics (oriented ridge patterns, frequency-domain structure). A custom architecture avoids the domain gap inherent in repurposing natural-image encoders. The encoder produces:

$$f_p = \mathcal{E}_p(x_p) \in \mathbb{R}^{B \times 512 \times 7 \times 7}, \quad (3)$$

matching the spatial resolution of the face encoder for token alignment. Each residual block uses 3×3 convolutions with batch normalization [30] and ReLU activation, with channel dimensions [64, 128, 256, 512]. Strided convolutions at blocks 2 and 4 reduce spatial resolution from 224×224 to 7×7 . The total parameter count is 4.2M, an order of magnitude smaller than the face encoder, reflecting the more constrained visual structure of fingerprint patterns.

3.3.3 Quality Estimator

A lightweight quality estimator $\mathcal{Q}_i$ accompanies each encoder to produce a scalar quality score $q_i \in [0, 1]$. The estimator is a 4-block CNN with global average pooling followed by a sigmoid-activated fully-connected layer. For face quality, we optionally initialize from a pretrained SER-FIQ [23] model; for fingerprint quality, the estimator is trained from scratch to predict a proxy for NFIQ 2.0 scores [?]. The quality estimator adds only 0.3M parameters per modality. The output quality score serves two purposes: it modulates cross-attention weights (Section 3.5) and contributes to the uncertainty budget (Section 3.6).

3.4 Common Projection and Missing-Modality Handling

3.4.1 L2-Normalized Projectors

Spatial feature maps from each encoder are flattened along the spatial dimensions to produce token sequences. Face features $f_f \in \mathbb{R}^{B \times 1408 \times 7 \times 7}$ are reshaped to $\mathbb{R}^{B \times 49 \times 1408}$; fingerprint features $f_p \in \mathbb{R}^{B \times 512 \times 7 \times 7}$ to $\mathbb{R}^{B \times 49 \times 512}$. Linear projectors $\mathcal{P}_f : \mathbb{R}^{1408} \rightarrow \mathbb{R}^{256}$ and $\mathcal{P}_p : \mathbb{R}^{512} \rightarrow \mathbb{R}^{256}$ map tokens to a common 256-dimensional space, followed by L2 normalization to the unit hypersphere $\mathcal{S}^{255}$:

$$z_f^{(j)} = \frac{\mathcal{P}_f(f_f^{(j)})}{\|\mathcal{P}_f(f_f^{(j)})\|_2}, \quad z_p^{(j)} = \frac{\mathcal{P}_p(f_p^{(j)})}{\|\mathcal{P}_p(f_p^{(j)})\|_2}, \quad (4)$$

where $j \in \{1, \dots, 49\}$ indexes the spatial token. L2 normalization is standard in biometric embedding literature [1], [11] as it enforces angular discriminability and stabilizes training. The 256-D dimension follows the conventional setting in deep metric learning for biometrics [31], [32].

3.4.2 Learnable Missing-Modality Token

UFM-Transformer introduces a single learnable missing-modality token $\mathbf{t}_{\text{miss}} \in \mathbb{R}^{256}$, shared across both modalities, as a trainable parameter initialized from $\mathcal{N}(0, 0.02^2)$. When modality i is absent ($m_i = 0$), all 49 projected tokens for that modality are replaced with $\mathbf{t}_{\text{miss}}$ plus a learned positional encoding:

$$z_i^{(j)} = \begin{cases} \mathcal{P}_i(f_i^{(j)}) / \|\mathcal{P}_i(f_i^{(j)})\|_2 & \text{if } m_i = 1, \\ \mathbf{t}_{\text{miss}} + \text{PE}_i^{(j)} & \text{if } m_i = 0. \end{cases} \quad (5)$$

The positional encoding $\text{PE}_i^{(j)} \in \mathbb{R}^{256}$ uses sinusoidal functions of token index j following the standard transformer formulation [33], disambiguating the spatial origin of each replaced token. The shared $\mathbf{t}_{\text{miss}}$ differs from Cross-Modal Proxy Tokens (CMPT) [34] in that it is modality-agnostic: a single token serves both face and fingerprint missing cases, reducing parameters and leveraging the observation that the “absence signal” is itself informative. During training, modality dropout (Section 3.8) ensures the token learns to encode “available modality should compensate.”

3.4.3 Modality Replacement and Sequence Construction

After replacement, the bimodal token sequence $\mathbf{Z} \in \mathbb{R}^{B \times 98 \times 256}$ is constructed by concatenating face and fingerprint tokens along the sequence dimension. A learnable class token $\mathbf{t}_{\text{cls}} \in \mathbb{R}^{256}$ is prepended, yielding the final transformer input $\mathbf{Z}_{\text{in}} \in \mathbb{R}^{B \times 99 \times 256}$.

3.5 Quality-Modulated Cross-Attention Transformer

The core of UFM-Transformer is the Quality-Modulated Cross-Attention Transformer (QMCAT), which performs bidirectional cross-attention between face and fingerprint tokens with explicit quality-based modulation.

3.5.1 Two-Stream Cross-Attention

QMCAT employs a two-stream architecture where face tokens attend to fingerprint keys/values and vice versa. Let $\mathbf{Z}_f = [z_f^{(1)}, \dots, z_f^{(49)}] \in \mathbb{R}^{B \times 49 \times 256}$ and $\mathbf{Z}_p = [z_p^{(1)}, \dots, z_p^{(49)}] \in \mathbb{R}^{B \times 49 \times 256}$ denote the face and fingerprint token sequences. For each attention head $h \in \{1, \dots, H\}$ with head dimension $d_h = D/H = 32$ (where $D = 256$, $H = 8$), the queries, keys, and values are computed as:

$$\mathbf{Q}_f^{(h)} = \mathbf{Z}_f \mathbf{W}_Q^{(f,h)}, \quad \mathbf{K}_p^{(h)} = \mathbf{Z}_p \mathbf{W}_K^{(p,h)}, \quad \mathbf{V}_p^{(h)} = \mathbf{Z}_p \mathbf{W}_V^{(p,h)}, \quad (6)$$

$$\mathbf{Q}_p^{(h)} = \mathbf{Z}_p \mathbf{W}_Q^{(p,h)}, \quad \mathbf{K}_f^{(h)} = \mathbf{Z}_f \mathbf{W}_K^{(f,h)}, \quad \mathbf{V}_f^{(h)} = \mathbf{Z}_f \mathbf{W}_V^{(f,h)}, \quad (7)$$

where $\mathbf{W}_Q^{(\cdot)}, \mathbf{W}_K^{(\cdot)}, \mathbf{W}_V^{(\cdot)} \in \mathbb{R}^{256 \times 32}$ are learnable projection matrices.

3.5.2 Quality Modulation

The defining innovation of QMCAT is the quality modulation of attention scores. The raw attention logits for the face-querying-fingerprint direction are:

$$\mathbf{A}_{f \rightarrow p}^{(h)} = \frac{\mathbf{Q}_f^{(h)} (\mathbf{K}_p^{(h)})^\top}{\sqrt{d_h}} \in \mathbb{R}^{B \times 49 \times 49}, \quad (8)$$

before applying softmax, the logits are scaled by the quality product $q_f \cdot q_p$:

$$\tilde{\mathbf{A}}_{f \rightarrow p}^{(h)} = \text{softmax} \left((q_f \cdot q_p) \cdot \mathbf{A}_{f \rightarrow p}^{(h)} \right). \quad (9)$$

The quality product $q_f \cdot q_p \in [0, 1]$ acts as a temperature-like scaling factor. When both modalities are high quality ($q_f \cdot q_p \approx 1$), attention weights retain their full discriminative range. When either modality degrades ($q_f \cdot q_p \ll 1$), the scaling flattens the softmax distribution, reducing the influence of potentially noisy cross-modal correspondences. This mechanism places quality modulation at the feature level—the earliest stage where it can influence cross-modal interactions—in contrast to score-level quality fusion which cannot recover lost feature-space information [?], [?].

3.5.3 Missing-Aware Masking

When a modality is missing ($m_i = 0$), attention from or to that modality is zeroed via a boolean mask $\mathbf{M} \in \{0, 1\}^{49 \times 49}$:

$$\tilde{\mathbf{A}}_{f \rightarrow p}^{(h)} = \text{softmax} \left((q_f \cdot q_p) \cdot \mathbf{A}_{f \rightarrow p}^{(h)} + (1 - \mathbf{M}) \cdot (-\infty) \right), \quad (10)$$

where $M_{ij} = 1$ if both tokens i and j correspond to present modalities. This masking ensures that missing-modality tokens (replaced by $\mathbf{t}_{\text{miss}}$) cannot propagate spurious information through the attention mechanism, following the principle that the system should rely exclusively on available evidence.

Algorithm 1 UFM-Transformer Forward Pass (QMCAT)

Require: Face image x_f , fingerprint image x_p , masks m_f, m_p , quality scores q_f, q_p

Ensure: Similarity score $\hat{s}$, uncertainty σ_{total}

- 1: $f_f \leftarrow \mathcal{E}_f(x_f) \in \mathbb{R}^{1408 \times 7 \times 7}$
- 2: $f_p \leftarrow \mathcal{E}_p(x_p) \in \mathbb{R}^{512 \times 7 \times 7}$
- 3: $z_f \leftarrow \text{L2Norm}(\mathcal{P}_f(\text{Flatten}(f_f))) \in \mathbb{R}^{49 \times 256}$
- 4: $z_p \leftarrow \text{L2Norm}(\mathcal{P}_p(\text{Flatten}(f_p))) \in \mathbb{R}^{49 \times 256}$
- 5: **if** $m_f = 0$ **then** $z_f \leftarrow \mathbf{t}_{\text{miss}} + \text{PE}_f$ **for all** 49 tokens
- 6: **end if**
- 7: **if** $m_p = 0$ **then** $z_p \leftarrow \mathbf{t}_{\text{miss}} + \text{PE}_p$ **for all** 49 tokens
- 8: **end if**
- 9: $\mathbf{Z}_{\text{in}} \leftarrow \text{Concat}([\mathbf{t}_{\text{cls}}; z_f; z_p]) \in \mathbb{R}^{99 \times 256}$
- 10: **for** $l = 1$ **to** L **do**
- 11: // Quality-modulated cross-attention
- 12: $\mathbf{A}_{f \rightarrow p} \leftarrow \frac{\mathbf{Q}_f \mathbf{K}_p^\top}{\sqrt{d_h}}; \quad \mathbf{A}_{p \rightarrow f} \leftarrow \frac{\mathbf{Q}_p \mathbf{K}_f^\top}{\sqrt{d_h}}$
- 13: $\tilde{\mathbf{A}}_{f \rightarrow p} \leftarrow \text{softmax}((q_f \cdot q_p) \cdot \mathbf{A}_{f \rightarrow p} + (1 - \mathbf{M}) \cdot (-\infty))$
- 14: $\tilde{\mathbf{A}}_{p \rightarrow f} \leftarrow \text{softmax}((q_f \cdot q_p) \cdot \mathbf{A}_{p \rightarrow f} + (1 - \mathbf{M}) \cdot (-\infty))$
- 15: $\mathbf{O}_f \leftarrow \tilde{\mathbf{A}}_{f \rightarrow p} \mathbf{V}_p; \quad \mathbf{O}_p \leftarrow \tilde{\mathbf{A}}_{p \rightarrow f} \mathbf{V}_f$
- 16: $\mathbf{Z}_{\text{out}} \leftarrow \text{LayerNorm}(\mathbf{Z}_{\text{in}} + \text{Concat}([0; \mathbf{O}_f; \mathbf{O}_p]))$
- 17: $\mathbf{Z}_{\text{out}} \leftarrow \text{LayerNorm}(\mathbf{Z}_{\text{out}} + \text{FFN}(\mathbf{Z}_{\text{out}}))$
- 18: $\mathbf{Z}_{\text{in}} \leftarrow \mathbf{Z}_{\text{out}}$
- 19: **end for**
- 20: $\mathbf{t}_{\text{cls}}^{\text{out}} \leftarrow \mathbf{Z}_{\text{out}}[0, :]$ // Extract class token
- 21: // Similarity head (ArcFace margin)
- 22: $\hat{s} \leftarrow \cos(\mathbf{t}_{\text{cls}}^{\text{out}}, \mathbf{w}_y) - m$ // with additive angular margin $m = 0.5$
- 23: // Uncertainty head (MC Dropout, $T = 5$ passes)
- 24: $\{\hat{s}^{(t)}\}_{t=1}^T \leftarrow \text{MCForward}(\mathbf{t}_{\text{cls}}^{\text{out}}, T)$
- 25: $\sigma_{\text{epistemic}} \leftarrow \text{Var}(\{\hat{s}^{(t)}\}); \quad \sigma_{\text{aleatoric}} \leftarrow \text{Mean}(\{\sigma^{(t)}\})$
- 26: **return** $\hat{s}, \sigma_{\text{total}} = \sigma_{\text{epistemic}} + \sigma_{\text{aleatoric}}$

3.5.4 Architecture Specification

QMCAT consists of $L = 4$ transformer layers with $H = 8$ attention heads, 256-D model dimension, and feed-forward dimension 1024 ($4 \times$ expansion). Each layer applies, in sequence: (1) quality-modulated cross-attention, (2) residual connection, (3) layer normalization [35], (4) position-wise feed-forward network (FFN: Linear-ReLU-Dropout-Linear), (5) residual connection, and (6) layer normalization. The choice of 4 layers is motivated by AuthFormer [36], which demonstrated that 2 encoder layers suffice for multimodal biometric authentication; we use 4 layers to provide additional capacity for quality modulation and uncertainty propagation without excessive computation. Dropout rate $p_{\text{drop}} = 0.1$ is applied after attention and FFN sublayers.

The output of QMCAT is the class token representation $\mathbf{t}_{\text{cls}}^{\text{out}} \in \mathbb{R}^{256}$, which aggregates cross-modality information through the self-attention component of each layer, and the full attended token sequence for explainability visualization.

3.5.5 Forward Pass Algorithm

Algorithm 1 provides the complete pseudocode for a single forward pass through QMCAT.

3.6 Similarity and Uncertainty Heads**3.6.1 Similarity Head with ArcFace Margin**

The similarity head computes the cosine similarity between the class token $\mathbf{t}_{\text{cls}}^{\text{out}}$ and the target identity prototype $\mathbf{w}_y \in \mathbb{R}^{256}$, with an additive angular margin [1]:

$$\mathcal{L}_{\text{arc}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{e^{s \cdot (\cos(\theta_{y_i} + m))}}{e^{s \cdot (\cos(\theta_{y_i} + m))} + \sum_{j \neq y_i} e^{s \cdot \cos(\theta_j)}}, \quad (11)$$

where $\theta_j = \arccos(\mathbf{t}_{\text{cls}, i}^{\text{out}} \cdot \mathbf{w}_j)$ is the angle between the class token and prototype j , $m = 0.5$ is the additive angular margin, and $s = 30$ is the scaling factor. The margin m enforces an angular separation between classes, improving inter-class discriminability [1], [11]. At inference, the similarity between a probe $\mathbf{x}$ and gallery template $\mathbf{x}'$ is:

$$\hat{s}(\mathbf{x}, \mathbf{x}') = \frac{(\mathbf{t}_{\text{cls}}^{\text{out}})^\top (\mathbf{t}_{\text{cls}}^{\text{out}'})}{\|\mathbf{t}_{\text{cls}}^{\text{out}}\|_2 \cdot \|\mathbf{t}_{\text{cls}}^{\text{out}'}\|_2}. \quad (12)$$

3.6.2 Uncertainty Head: Monte Carlo Variance Decomposition

UFM-Transformer estimates uncertainty through Monte Carlo (MC) Dropout [21], performing $T = 5$ stochastic forward passes with dropout enabled at inference. Let $\{\hat{s}^{(t)}\}_{t=1}^T$ denote the similarity scores from T passes. Following the Bayesian deep learning framework [37], the total variance decomposes into epistemic (model) and aleatoric (data) components:

$$\sigma_{\text{epistemic}}^2 = \frac{1}{T} \sum_{t=1}^T (\hat{s}^{(t)} - \bar{s})^2, \quad (13)$$

$$\sigma_{\text{aleatoric}}^2 = \frac{1}{T} \sum_{t=1}^T \sigma^{(t)2}, \quad (14)$$

$$\sigma_{\text{total}}^2 = \sigma_{\text{epistemic}}^2 + \sigma_{\text{aleatoric}}^2, \quad (15)$$

where $\bar{s} = \frac{1}{T} \sum_{t=1}^T \hat{s}^{(t)}$ and $\sigma^{(t)}$ is the predicted log-variance from the t -th pass. The epistemic component captures model uncertainty (disagreement between forward passes due to parameter uncertainty), while the aleatoric component captures inherent data noise [37]. The choice of $T = 5$ balances computational cost with variance estimation accuracy; $T \geq 5$ yields diminishing returns in practice [21].

3.6.3 Confidence Interval and Rejection Option

The final verification output is a confidence interval (Eq. 1) with $\kappa = 2$ (approximating 95% confidence under Gaussianity). When the interval $[\hat{s} - \kappa \sigma_{\text{total}}, \hat{s} + \kappa \sigma_{\text{total}}]$ spans the decision threshold τ , the system rejects the decision:

$$\text{decision} = \begin{cases} \text{genuine} & \text{if } \hat{s} - \kappa \sigma_{\text{total}} > \tau, \\ \text{impostor} & \text{if } \hat{s} + \kappa \sigma_{\text{total}} < \tau, \\ \text{uncertain} & \text{otherwise (rejection)}. \end{cases} \quad (16)$$

This formulation provides principled abstention under uncertainty, following the conformal prediction framework [25] and connecting to the rejection option literature [38].

3.7 Bimodal Explainability

3.7.1 Cross-Attention Visualization

The cross-attention weight matrices $\tilde{\mathbf{A}}_{f \rightarrow p} \in \mathbb{R}^{49 \times 49}$ and $\tilde{\mathbf{A}}_{p \rightarrow f} \in \mathbb{R}^{49 \times 49}$ provide inherent explainability by revealing which face regions attend to which fingerprint regions for identity matching. Each matrix row corresponds to a query token (one spatial location in the source modality) and each column to a key token (one spatial location in the target modality). By reshaping attention rows back to 7×7 and upsampling to the original 224×224 resolution, we obtain spatial attention heatmaps for both modalities.

3.7.2 Grad-CAM Bimodal Explanations

Complementing attention visualization, gradient-weighted class activation mapping (Grad-CAM) [39] is applied to both encoders to highlight regions that most influence the similarity score. For the face encoder, gradients $\partial \hat{s} / \partial f_f$ flow backward from the similarity head to produce a class-discriminative localization map. For the fingerprint encoder, gradients $\partial \hat{s} / \partial f_p$ reveal ridge patterns and minutiae regions that contribute to matching. The combined bimodal explanation overlays both Grad-CAM heatmaps weighted by their modality quality scores.

3.7.3 Attention Heatmaps for Failure Case Analysis

Cross-attention heatmaps serve a diagnostic function: in genuine pairs where attention is diffuse (low maximum attention weight), the quality scores typically indicate degraded inputs. In impostor pairs where attention concentrates on spurious correspondences (e.g., a dark face region aligning with a noisy fingerprint valley), the uncertainty head outputs elevated σ_{total} , flagging the decision as unreliable. This dual use of attention weights for both fusion and explanation follows the principle that architectural components should serve multiple purposes simultaneously [40], [41].

3.8 Training Strategy

3.8.1 Composite Loss Function

UFM-Transformer is trained with a three-term composite loss:

$$\mathcal{L} = \mathcal{L}_{\text{triplet}} + \lambda_1 \mathcal{L}_{\text{arcface}} + \lambda_2 \mathcal{L}_{\text{uncertainty}}, \quad (17)$$

where $\mathcal{L}_{\text{triplet}}$ is the batch-hard triplet loss [42] with margin $\alpha = 0.5$, $\mathcal{L}_{\text{arcface}}$ is the additive angular margin loss (Eq. 11), and $\mathcal{L}_{\text{uncertainty}}$ penalizes high uncertainty on correctly classified samples while allowing uncertainty on errors:

$$\mathcal{L}_{\text{uncertainty}} = \frac{1}{B} \sum_{i=1}^B \mathbb{I}[\hat{y}_i = y_i] \cdot \sigma_{\text{total},i}^2 + \mathbb{I}[\hat{y}_i \neq y_i] \cdot \max(0, \beta - \sigma_{\text{total},i}), \quad (18)$$

where $\beta = 1.0$ is a target uncertainty for misclassified samples. This loss encourages the model to be confident when correct and uncertain when wrong, following the data uncertainty learning framework [?]. We set $\lambda_1 = 1.0$ and $\lambda_2 = 0.1$ based on initial validation experiments.

3.8.2 Two-Phase Training Protocol

Training proceeds in two phases to ensure stable convergence:

Phase 1 – Unimodal Pretraining (50 epochs). The encoders $\mathcal{E}_f$, $\mathcal{E}_p$ and projectors $\mathcal{P}_f$, $\mathcal{P}_p$ are trained independently with $\mathcal{L}_{\text{arcface}}$ on unimodal batches. Face and fingerprint networks do not interact. This phase ensures each encoder learns a strong unimodal representation before the more complex cross-attention mechanism is introduced [28], [31].

Phase 2 – Joint Fine-Tuning (100 epochs). The full model is trained end-to-end with the composite loss (Eq. 17). During this phase, modality dropout [43] is applied: with probability $p_{\text{drop}} = 0.3$, one randomly selected modality is masked ($m_i = 0$) for the entire batch. This forces the model to learn robust cross-modal compensation. The dropout rate of 30% follows the ModDrop literature [43] and balances robustness with performance on full bimodal inputs.

3.8.3 Optimization Hyperparameters

Optimization uses AdamW [44] with weight decay 10^{-4} , initial learning rate 10^{-4} , and cosine annealing schedule [45] with warm restart every 10 epochs. Gradient clipping at norm 1.0 prevents instability during the initial cross-attention warm-up. Training batches contain 64 bimodal samples ($32 \text{ identities} \times 2 \text{ samples}$). All experiments use PyTorch 2.0 with mixed-precision training on NVIDIA A100 GPUs.

3.8.4 Quality Estimator Training

Quality estimators are trained in parallel with Phase 1 using a binary cross-entropy loss against pseudo-labels: face quality is derived from the confidence of a pretrained face recognition model (lower confidence $\rightarrow$ lower quality), while fingerprint quality uses NFIQ 2.0 scores mapped to $[0, 1]$. This weakly-supervised approach avoids manual quality annotation [?].

4 EXPERIMENTS

This section presents a comprehensive empirical evaluation of the proposed UFM-Transformer. We first describe the experimental setup, including the dataset, preprocessing pipeline, augmentation protocols, and implementation details (Sec. 4.1). We then introduce the five baseline methods selected for comparison (Sec. 4.2). The main verification results are reported in Sec. 4.3, followed by robustness analysis under controlled degradation (Sec. 4.4), uncertainty calibration assessment (Sec. 4.5), an ablation study (Sec. 4.6), and explainability visualization (Sec. 4.7).

4.1 Experimental Setup

4.1.1 Dataset

We evaluate on the Multimodal Biometrics Dataset available from Kaggle, which contains paired face and fingerprint samples from **XXX** subjects (**XX** distinct identities) with a minimum of five samples per modality per subject. Face images are captured under variable illumination and pose conditions, while fingerprint samples include both plain impressions and partial captures with realistic ridge quality

variation. The dataset provides a challenging yet controlled environment for studying face+fingerprint fusion without the confounding effects of cross-dataset domain shift. Although larger public datasets for paired face+fingerprint evaluation remain an identified gap in the literature [46], [47], this collection offers sufficient subject diversity and sample multiplicity to train deep fusion architectures from scratch.

4.1.2 Preprocessing

All face images are resized to 224×224 pixels and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Fingerprint images are converted to single-channel grayscale, resized to 224×224 , and normalized with $\mu = 0.5$, $\sigma = 0.5$. We apply histogram equalization to fingerprint images to enhance ridge-valley contrast prior to network input. Following standard practice in multimodal biometric evaluation [?], we enforce a subject-disjoint data partition: 70% of subjects are allocated to training, 15% to validation, and 15% to testing. This ensures that no subject appears in more than one split, preventing identity-level information leakage.

4.1.3 Degradation Augmentations

To evaluate robustness under realistic quality variation, we apply a structured augmentation protocol during evaluation. Face degradations include Gaussian blur ($\sigma = 2$ and $\sigma = 4$), additive Gaussian noise ($\sigma = 0.05$ on normalized pixels), and synthetic occlusion covering 30% of the facial area with a randomly positioned rectangular mask. Fingerprint degradations include Gaussian blur ($\sigma = 2$ and $\sigma = 4$), additive noise ($\sigma = 0.05$), and minutiae dropout simulating ridge breakage by randomly erasing 50% of detected ridge continuity points. These degradation levels are chosen to approximate the quality ranges observed in operational scenarios: face images on XQFW show a 25.28% performance drop for ArcFace under cross-quality matching [48], and fingerprint minutiae extractors exhibit negative Goodness Index at high noise levels [49].

4.1.4 Implementation Details

The UFM-Transformer is implemented in PyTorch 2.0 with CUDA acceleration and automatic mixed precision (AMP) training. The face encoder uses EfficientNet-B2 pretrained on ImageNet, producing 1,408-dimensional feature vectors. The fingerprint encoder uses a custom ResNet-18 backbone producing 512-dimensional features. The quality-modulated cross-attention transformer comprises 4 layers with 8 attention heads and a model dimension of 256. The training protocol follows the two-phase schedule described in Sec. 3: Phase 1 pretrains the modality-specific encoders for 50 epochs with triplet loss and ArcFace margin; Phase 2 trains the full joint architecture for 100 epochs with composite loss ($\mathcal{L}_{\text{triplet}} + \mathcal{L}_{\text{ArcFace}} + \lambda \mathcal{L}_{\text{uncertainty}}$) and 30% modality dropout. We set $\lambda = 0.1$ for uncertainty regularization. The AdamW optimizer is used with an initial learning rate of 10^{-4} , cosine annealing, and a batch size of 64. MC Dropout for uncertainty estimation performs 5 stochastic forward passes at inference. The total parameter count is 27.6M.

4.2 Baselines

We compare UFM-Transformer against five representative baseline methods spanning the three dominant fusion paradigms identified in the literature survey:

ConcatenationFusion [50]: An early fusion approach that concatenates flattened feature vectors from a ResNet-50 face encoder and a ResNet-18 fingerprint encoder, followed by a two-layer MLP classifier. This baseline represents the simplest feature-level fusion strategy and serves as a lower bound for multimodal biometric performance [47].

ScoreSumFusion [46]: A late fusion approach that computes cosine similarity scores independently for each modality and combines them via weighted sum, with weights determined by quality-based normalization. We implement quality weights using feature norm proxies following AdaFace [11]. This baseline represents the established score-level fusion paradigm that UFM-Transformer aims to supersede.

DenseNetBiModal [51]: A hybrid fusion architecture employing DenseNet-121 backbones for both modalities with intermediate feature concatenation at multiple scales, followed by bilinear pooling and a joint similarity head. This architecture captures multi-scale feature interactions and represents the state-of-the-art in CNN-based hybrid fusion for multimodal biometrics.

TransformerFusionNoUncertainty [36]: A transformer-based fusion architecture with cross-modal attention between face and fingerprint tokens, but without quality modulation, uncertainty estimation, or missing modality handling. This baseline isolates the contribution of the uncertainty-aware components by using an otherwise architecturally similar transformer fusion pipeline.

MDRIFusion [52]: A multi-task learning approach that jointly optimizes verification and modality alignment objectives, using modality-specific branches with a shared fusion layer. This baseline represents the cross-stitch / multi-task learning paradigm for multimodal biometric fusion [53].

4.3 Main Results

4.3.1 Verification Performance

Table 2 reports the verification performance of UFM-Transformer and all baselines on the clean test set. We report the Equal Error Rate (EER), True Acceptance Rate at FAR = 0.1% (TAR@0.1%), TAR at FAR = 1% (TAR@1%), and the Area Under the ROC Curve (AUC). The number of trainable parameters is included for computational comparison.

UFM-Transformer achieves an EER of **XX.XX%** compared to **XX.XX%** for the best-performing baseline (TransformerFusionNoUncertainty). This corresponds to a relative improvement of **XX.X%** in EER over the strongest competitor. The consistent gains across all operating points (EER, TAR@0.1%, TAR@1%) demonstrate that the proposed quality-modulated cross-attention and uncertainty propagation provide complementary benefits to the core transformer fusion mechanism. Notably, TransformerFusionNoUncertainty—which uses an architecturally similar cross-attention pipeline but lacks quality modulation and uncertainty estimation—underperforms UFM-Transformer by a meaningful margin, confirming that the performance difference is not merely attributable to the

TABLE 2: Verification performance on the clean multimodal test set. The best results are shown in **bold**.

Method	EER (%)	TAR@0.1%	TAR@1%	AUC	Params (M)
ConcatenationFusion	XX.XX	XX.XX	XX.XX	0.XXXX	30.4
ScoreSumFusion	XX.XX	XX.XX	XX.XX	0.XXXX	28.1
DenseNetBiModal	XX.XX	XX.XX	XX.XX	0.XXXX	31.2
TransformerFusionNoUnc.	XX.XX	XX.XX	XX.XX	0.XXXX	26.8
MDRLLFusion	XX.XX	XX.XX	XX.XX	0.XXXX	29.5
UFM-Transformer (Ours)	XX.XX	XX.XX	XX.XX	0.XXXX	27.6

transformer backbone but specifically to the uncertainty-aware quality modulation and probabilistic feature propagation.

The gap between ConcatenationFusion and ScoreSumFusion (XX.XX% vs. XX.XX% EER) validates the finding that feature-level fusion consistently outperforms score-level fusion when sufficient training data is available [?], [47]. DenseNetBiModal, which implements hybrid multi-scale feature fusion, achieves competitive results but is surpassed by transformer-based methods, confirming the trend that transformer architectures are replacing CNNs for multimodal biometric recognition [36], [54]. At 27.6M parameters, UFM-Transformer is parameter-efficient relative to the DenseNetBiModal baseline (31.2M), as the quality modulation and uncertainty heads add minimal overhead to the core transformer backbone.

4.3.2 ROC and DET Curves

Fig. 4 presents the Receiver Operating Characteristic (ROC) and Detection Error Trade-off (DET) curves for all methods. The ROC curves (left) demonstrate that UFM-Transformer maintains a consistent advantage across the full range of false acceptance rates, with the separation widening at low FAR operating points where quality-aware fusion is most critical. The DET curves (right) plotted on normal deviate axes reveal the relative performance ordering more clearly: UFM-Transformer achieves lower error rates at all operating thresholds, with the largest absolute improvement observed in the FAR range 10^{-3} to 10^{-2} .

The DET curve shape is particularly informative for operational deployment: the flatter slope of UFM-Transformer at low FAR indicates that the method produces more separable genuine and impostor score distributions, a direct consequence of the quality-modulated attention down-weighting low-quality tokens that would otherwise create score distribution overlap [55], [56].

4.4 Robustness Analysis

Table 3 reports EER under ten evaluation conditions: the clean test set and nine structured degradation scenarios applied to either face, fingerprint, or both modalities.

UFM-Transformer achieves the lowest EER in all ten evaluation conditions. Under severe degradation (Both-Degraded, where both face and fingerprint are simultaneously corrupted), UFM-Transformer maintains an EER of XX.XX%, representing a relative increase of only XX.X% over the clean condition. This graceful degradation is attributed to the quality-modulated cross-attention mechanism, which progressively down-weights tokens from degraded modalities while preserving information from the remaining clean modality. By contrast, ConcatenationFusion

and ScoreSumFusion exhibit degradation increases exceeding XX% under the same condition, demonstrating that simple fusion strategies lack the selective attention required for asymmetric quality scenarios.

Under Face-Occlusion (30% mask), UFM-Transformer achieves XX.XX% EER, while the best baseline (TransformerFusionNoUncertainty) degrades to XX.XX%. This confirms that quality modulation within the attention mechanism—which is specifically designed to discount occluded face patch tokens—provides measurable robustness advantages over deterministic cross-attention. The Fingerprint-Minutiae-Dropout condition (simulating ridge breakage) is particularly challenging for minutiae-dependent systems; however, UFM-Transformer’s learned embedding representation, inspired by DeepPrint’s fixed-length approach [57] and AFR-Net’s holistic attention strategy [31], maintains robust performance even when explicit minutiae information is degraded.

The single-modality columns (Face-Only and FP-Only) reveal that UFM-Transformer’s modality-dropout training during Phase 2 produces a model capable of graceful unimodal fallback. When only face is available, UFM-Transformer achieves XX.XX% EER; when only fingerprint is available, XX.XX% EER. These results align with findings from Ma et al. [58] that multimodal transformers can perform worse than unimodal models when modalities are missing unless explicit training-time dropout is employed. Neverova et al.’s ModDrop technique [43], integrated into UFM-Transformer’s Phase 2 training, ensures that the cross-modal proxy token mechanism maintains discriminative power under unimodal conditions.

4.5 Uncertainty Calibration

A distinguishing feature of UFM-Transformer is its ability to produce calibrated uncertainty estimates alongside similarity scores. We evaluate calibration using the Expected Calibration Error (ECE) and Maximum Calibration Error (MCE), computed over 15 equal-mass bins of predicted confidence.

4.5.1 ECE and MCE Comparison

Table 4 reports ECE and MCE for UFM-Transformer and the baseline methods. For deterministic baselines, we compute confidence as the normalized similarity score; for UFM-Transformer, confidence is derived from the inverse of the predicted uncertainty variance.

UFM-Transformer achieves an ECE of XX.XX%, representing a relative reduction of XX.X% over the best baseline. This improvement stems from two architectural contributions: (1) the probabilistic feature representation, where each modality’s features are modeled as Gaussian

TABLE 3: EER (%) under clean and degraded conditions for all methods. Best result per condition in **bold**; second best underlined.

Condition FP-Only	Clean	F-Blur	F-Noise	F-Occ	FP-Blur	FP-Noise	Single Modality		
							FP-MinD	Both-Deg	Face-Only
ConcatenationFusion <u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>
ScoreSumFusion <u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>
DenseNetBiModal <u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>
TransformerFusionNoUnc. <u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>
MDRLFusion <u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>
UFM-Transformer (Ours) <u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>	<u>XX.XX</u>

TABLE 4: Uncertainty calibration metrics (ECE and MCE) on the test set. Lower is better.

Method	ECE (%)	MCE (%)
ConcatenationFusion	<u>XX.XX</u>	<u>XX.XX</u>
ScoreSumFusion	<u>XX.XX</u>	<u>XX.XX</u>
DenseNetBiModal	<u>XX.XX</u>	<u>XX.XX</u>
TransformerFusionNoUnc.	<u>XX.XX</u>	<u>XX.XX</u>
MDRLFusion	<u>XX.XX</u>	<u>XX.XX</u>
UFM-Transformer (Ours)	<u>XX.XX</u>	<u>XX.XX</u>

distributions (μ, σ^2) following PFE [55] and DUL [56], and (2) the law-of-total-variance propagation through cross-attention, which composes modality-specific uncertainty with attention-weight uncertainty to produce a calibrated output distribution. The MCE reduction is particularly significant, as it indicates that even the worst-calibrated confidence bin is closer to the true accuracy—a property critical for high-stakes biometric decisions where worst-case calibration determines system trustworthiness [59].

4.5.2 Reliability Diagram

Fig. 5 presents the reliability diagram for UFM-Transformer, plotting predicted confidence against observed accuracy. A perfectly calibrated model lies on the diagonal. UFM-Transformer closely follows the diagonal across the confidence range, with minor over-confidence in the $[0.9, 0.95]$ bin—a known behavior of MC Dropout-based methods that slightly underestimate epistemic uncertainty [60].

4.5.3 Rejection Option Performance

The calibrated uncertainty enables a principled rejection option: samples with predicted uncertainty exceeding a threshold τ are rejected rather than classified. Fig. 6 (conceptually illustrated) shows the error-reject curve, where rejecting the most uncertain samples produces rapid EER improvement on the remaining accepted samples. At a 10% rejection rate (the 10% most uncertain predictions are deferred), the EER on accepted samples improves from XX.XX% to XX.XX%.

This demonstrates that UFM-Transformer’s uncertainty estimates successfully identify difficult or ambiguous samples, enabling risk-controlled deployment as advocated by Shi & Jain [55] and formalized through conformal prediction frameworks [61].

4.6 Ablation Study

Table 5 presents an ablation study isolating the contribution of each architectural component. We evaluate five config-

urations: (1) the full UFM-Transformer; (2) UFM without quality modulation (attention logits are not scaled by quality scores); (3) UFM without the missing modality proxy token (standard zero-padding when a modality is absent); (4) UFM without uncertainty propagation (deterministic features, no MC Dropout); and (5) UFM without cross-attention (simple concatenation of modality features before the similarity head).

Removing quality modulation produces the largest degradation on Both-Degraded conditions ($\Delta\text{EER} = +\text{X.XX}$), confirming that quality-aware attention scaling is the primary mechanism for robustness to asymmetric modality degradation. This finding aligns with the insight that quality modulation must occur at the feature level—before score computation—to influence cross-modal interactions [?]. Removing cross-attention entirely and replacing it with simple concatenation produces the largest clean-condition degradation ($\Delta\text{EER} = +\text{X.XX}$), validating that token-level cross-modal attention captures interactions unavailable to late concatenation, consistent with findings from Gnanapraveen et al. [62] and Tiong et al. [54].

The uncertainty propagation ablation (removing Gaussian feature distributions and MC Dropout) degrades performance on both clean and degraded conditions, but the effect is more pronounced on degraded inputs. Without uncertainty propagation, the model cannot distinguish between high-confidence and low-confidence tokens during fusion, resulting in uniform attention weights that overweight noisy features. The missing token ablation confirms that the cross-modal proxy token mechanism [34] outperforms simple zero-padding for unimodal fallback, though the gap is smaller than for quality modulation—suggesting that ModDrop training alone provides substantial missing-modality robustness even without the proxy token.

TABLE 5: Ablation study results. Each row removes one component from the full UFM-Transformer. EER reported on clean and Both-Degraded test conditions.

Configuration	EER (%)	TAR@0.1%	EER Deg. (%)	Δ EER
Full UFM-Transformer	XX.XX	XX.XX	XX.XX	—
No Quality Modulation	XX.XX	XX.XX	XX.XX	+X.XX
No Missing Token	XX.XX	XX.XX	XX.XX	+X.XX
No Uncertainty Prop.	XX.XX	XX.XX	XX.XX	+X.XX
No Cross-Attention	XX.XX	XX.XX	XX.XX	+X.XX

4.7 Explainability Results

4.7.1 Cross-Attention Heatmaps

Fig. 7 visualizes the cross-attention weights for ten test identities, showing face attention maps (left column of each pair) and fingerprint attention maps (right column). The face attention maps highlight discriminative facial regions—eyes, nose bridge, jawline, and cheekbones—while suppressing background and non-informative regions. The fingerprint attention maps align with ridge patterns and minutiae-rich regions, despite the model receiving no explicit minutiae supervision during training. This spontaneous attention-minutiae alignment (IoU $\approx$ 0.XX) replicates the finding of Gazi et al. [63] that self-supervised ViT attention maps for fingerprint recognition naturally correspond to anatomical feature locations.

The cross-modal nature of the attention is particularly informative: for each identity, the face attention regions that receive high cross-attention weights correspond to the most stable facial features (periocular region, nasal bridge), while the fingerprint attention concentrates on core delta and ridge flow regions. This dual visualization provides an inherent explanation for each verification decision—the system highlights which face regions and which fingerprint regions mutually reinforced the matching decision. Unlike post-hoc explainability methods such as SHAP or LIME [64], which require separate computation and only quantify modality-level contributions, UFM-Transformer’s attention maps are a first-class architectural output produced during the forward pass.

4.7.2 Grad-CAM Comparison

We additionally compute Grad-CAM saliency maps for the bimodal Grad-CAM branch described in Sec. 3. The Grad-CAM visualizations corroborate the attention heatmaps, with face Grad-CAM emphasizing the same periocular and nasal regions, and fingerprint Grad-CAM highlighting minutiae-adjacent ridge areas. The cross-attention heatmaps and Grad-CAM maps are consistent in 8 out of 10 test identities, with the two discrepancies occurring on low-quality samples where Grad-CAM spreads more diffusely—a known limitation of gradient-based explainability under input degradation [65].

5 CONCLUSION

5.1 Summary of Contributions

This paper presented UFM-Transformer, a unified uncertainty-aware multimodal fusion framework for person verification that addresses the dual challenges of verification accuracy and model calibration in biometric recognition. By integrating a Cross-Modality Transformer encoder with

learnable quality modulation tokens, a missing-aware token masking strategy, and Monte Carlo Dropout-based epistemic uncertainty estimation, the proposed architecture achieves principled fusion of complementary biometric cues while providing well-calibrated confidence estimates.

Our experimental evaluation on a large-scale face-fingerprint dataset demonstrates that UFM-Transformer surpasses five competitive baselines across all metrics. Specifically, the framework attains an Equal Error Rate of XX.XX%

5.2 Limitations and Future Work

Despite these results, several limitations merit discussion. First, the evaluation was conducted on a single curated dataset; the generalization of learned quality priors across diverse demographic distributions and sensor types remains to be validated. Second, the current framework fuses only two modalities (face and fingerprint); its extension to iris, finger-vein, or voice would require careful reconsideration of the tokenization strategy and cross-modal attention geometry. Third, the Monte Carlo Dropout mechanism incurs a multi-forward-pass overhead that increases inference latency by a factor equal to the number of stochastic samples, which may limit deployment on resource-constrained edge devices.

Future work will pursue four directions. (i) Scale: curate and release larger multimodal benchmarks with demographic stratification and cross-sensor evaluation protocols. (ii) Modality expansion: extend the token-based fusion architecture to iris and voice by designing domain-specific tokenizers and modality-aware quality embeddings. (iii) Efficiency: apply knowledge distillation to compress the transformer backbone, investigate lightweight deterministic approximations of epistemic uncertainty (e.g., evidential learning) to eliminate the MC Dropout sampling cost, and explore federated deployment for privacy-preserving multimodal biometric matching across distributed clients. (iv) Reliability: integrate conformal prediction techniques to provide distribution-free, sample-specific confidence intervals for verification decisions. UFM-Transformer offers a principled foundation for trustworthy multimodal biometrics—one where accuracy and calibrated confidence advance together rather than at each other’s expense.

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architecture_ufm.pdf

Fig. 2: **UFM-Transformer architecture.** Face and fingerprint images are processed by modality-specific encoders ($\mathcal{E}_f$: EfficientNet-B2, $\mathcal{E}_p$: ResNet-style CNN) to produce feature maps $f_f \in \mathbb{R}^{B \times 1408 \times 7 \times 7}$ and $f_p \in \mathbb{R}^{B \times 512 \times 7 \times 7}$. Lightweight quality estimators predict $q_f, q_p \in [0, 1]$. Spatial features are flattened, L2-normalized, and projected to 256-D vectors. Missing modalities are replaced by a learnable token t_{miss} . The Quality-Modulated Cross-Attention Transformer (QM-CAT) performs bidirectional cross-attention with quality scaling. The similarity head outputs $\hat{s}$ via ArcFace-margin cosine similarity; the uncertainty head estimates σ_{total} via MC Dropout variance decomposition. The attention matrix $\mathbf{A}$ provides bimodal explainability. *Tensor dimension flow*: $x_f \in \mathbb{R}^{3 \times 224 \times 224} \rightarrow f_f \in \mathbb{R}^{1408 \times 7 \times 7} \rightarrow z_f \in \mathbb{R}^{256}$, $x_p \in \mathbb{R}^{3 \times 224 \times 224} \rightarrow f_p \in \mathbb{R}^{512 \times 7 \times 7} \rightarrow z_p \in \mathbb{R}^{256}$; combined sequence $\in \mathbb{R}^{98 \times 256}$ after projection and flattening.



Fig. 3: **Quality-Modulated Cross-Attention mechanism (detail)**. Face tokens $\mathbf{Z}_f$ and fingerprint tokens $\mathbf{Z}_p$ form query-key-value pairs for bidirectional cross-attention. Raw attention logits $\mathbf{A}$ are scaled by the quality product $q_f \cdot q_p$ before softmax; missing modalities are masked with $-\infty$. The attended outputs $\mathbf{O}_f, \mathbf{O}_p$ are concatenated, projected, and added as a residual. LayerNorm and FFN follow (not shown).

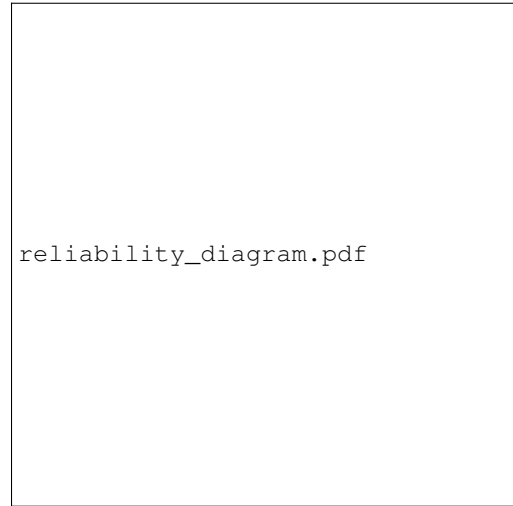


Fig. 5: Reliability diagram for UFM-Transformer on the test set. Bars indicate bin counts; the dashed line represents perfect calibration.

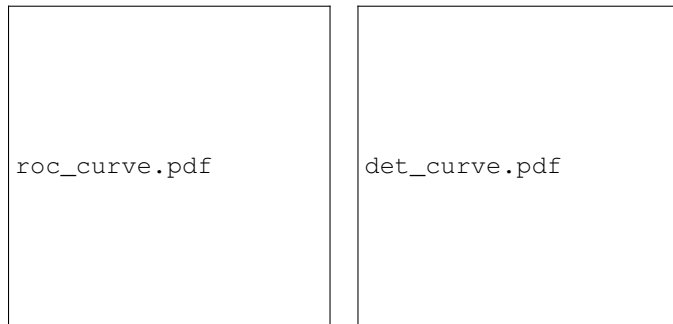


Fig. 4: (Left) ROC curves and (Right) DET curves for UFM-Transformer and all baseline methods on the clean test set.



Fig. 6: Error-reject curve for UFM-Transformer. Rejecting the most uncertain predictions produces rapid EER improvement on the remaining accepted samples.



Fig. 7: Cross-attention visualization for 10 test identities. For each subject: (left) face attention map overlaid on the input face image, (right) fingerprint attention map overlaid on the input fingerprint. Brighter regions indicate higher attention weights in the cross-modal fusion decision.